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java.nio.channels.spi

Class AsynchronousChannelProvider

java.lang.Object
    java.nio.channels.spi.AsynchronousChannelProvider

```
public abstract class AsynchronousChannelProvider
extends Object
```

Service-provider class for asynchronous channels.

An asynchronous channel provider is a concrete subclass of this class that has a zero-argument constructor and implements the abstract methods specified below. A given invocation of the Java virtual machine maintains a single system-wide default provider instance, which is returned by the `provider` method. The first invocation of that method will locate the default provider as specified below.

All of the methods in this class are safe for use by multiple concurrent threads.

Since:

1.7

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	`AsynchronousChannelProvider()`	Initializes a new instance of this class.

Method Summary

Concrete

Modifier and Type	Method
abstract **AsynchronousChannelGroup**	**openAsynchronousChannelGroup**(**ExecutorSe** int initialSize)
abstract **AsynchronousChannelGroup**	**openAsynchronousChannelGroup**(int nThrea **ThreadFactory** threadFactory)
abstract **AsynchronousServerSocketChannel**	**openAsynchronousServerSocketChannel**(**Asy**
abstract **AsynchronousSocketChannel**	**openAsynchronousSocketChannel**(**Asynchron**
static **AsynchronousChannelProvider**	**provider**()

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

AsynchronousChannelProvider

```
protected AsynchronousChannelProvider()
```

Initializes a new instance of this class.

Throws:

`SecurityException` - If a security manager has been installed and it denies `RuntimePermission("asynchronousChannelProvider")`

Method Detail

provider

```
public static AsynchronousChannelProvider provider()
```

Returns the system-wide default asynchronous channel provider for this invocation of the Java virtual machine.

The first invocation of this method locates the default provider object as follows:

1. If the system property `java.nio.channels.spi.AsynchronousChannelProvider` is defined then it is taken to be the fully-qualified name of a concrete provider class. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

2. If a provider class has been installed in a jar file that is visible to the system class loader, and that jar file contains a provider-configuration file named `java.nio.channels.spi.AsynchronousChannelProvider` in the resource directory `META-INF/services`, then the first class name specified in that file is taken. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

3. Finally, if no provider has been specified by any of the above means then the system-default provider class is instantiated and the result is returned.

Subsequent invocations of this method return the provider that was returned by the first invocation.

Returns:

The system-wide default AsynchronousChannel provider

openAsynchronousChannelGroup

```
public abstract AsynchronousChannelGroup openAsynchronousChannelGroup(int nThread
                                                                      ThreadFacto
                                                                    throws IOExceptior
```

Constructs a new asynchronous channel group with a fixed thread pool.

Parameters:

nThreads - The number of threads in the pool

threadFactory - The factory to use when creating new threads

Returns:

A new asynchronous channel group

Throws:

IllegalArgumentException - If nThreads <= 0

IOException - If an I/O error occurs

See Also:

AsynchronousChannelGroup.withFixedThreadPool(int, java.util.concurrent.ThreadFactory)

openAsynchronousChannelGroup

```
public abstract AsynchronousChannelGroup openAsynchronousChannelGroup(ExecutorSer
                                                                      int initial
                                                                    throws IOExceptior
```

Constructs a new asynchronous channel group with the given thread pool.

Parameters:

executor - The thread pool

initialSize - A value >=0 or a negative value for implementation specific default

Returns:

A new asynchronous channel group

Throws:

IOException - If an I/O error occurs

See Also:

AsynchronousChannelGroup.withCachedThreadPool(java.util.concurrent.ExecutorServic
int)

openAsynchronousServerSocketChannel

```
public abstract AsynchronousServerSocketChannel openAsynchronousServerSocketChann
                                                                                thro
```

Opens an asynchronous server-socket channel.

Parameters:

group - The group to which the channel is bound, or null to bind to the default group

Returns:

The new channel

Throws:

IllegalChannelGroupException - If the provider that created the group differs from this provider

ShutdownChannelGroupException - The group is shutdown

IOException - If an I/O error occurs

openAsynchronousSocketChannel

```
public abstract AsynchronousSocketChannel openAsynchronousSocketChannel(Asynchron
                                                                        throws IOExcepti
```

Opens an asynchronous socket channel.

Parameters:

group - The group to which the channel is bound, or null to bind to the default group

Returns:

The new channel

Throws:

IllegalChannelGroupException - If the provider that created the group differs from this provider

ShutdownChannelGroupException - The group is shutdown

```
IOException - If an I/O error occurs
```

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Submit a bug or feature
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